2	**(a)**		From Newton's third law, if body A exerts a force on body B, then body B will exert a force of the same type that is equal in magnitude and opposite in direction on body A. Hence, $(ma)_A = -(ma)_B$, where m represents mass and a represents acceleration By Newton's second law, $m_A(v_A-u_A)/t = - m_B(v_B-u_B)/t$ For the same duration of interaction, the gain in momentum by body A is equal to the lost in momentum by body B. This shows principle of conservation of momentum. Comments: Stating the definitions of either Newton's second law, third law and Conservation of momentum (COM) without showing how they are inter-related which leads to the deduction of COM will not earn any credit. Newton's third law is **NOT** stated as "for every action there is an equal and opposite reaction"!	M1 M1
	(b)	**(i)**	$mv_0 + 0 = m(0.5\ v_0)$ + momentum of nucleus at t_1 momentum of nucleus at $t_1 = 0.5\ mv_0$	B1
		(ii)	By conservation of momentum, total momentum of the system is conserved since no external force acts on the system. $\therefore$ total initial momentum of the system = total momentum of the system at t_1.	B1 A1

Comments:
Students must realise that in order for COM to be applied, the premise is there is no resultant external force acting on the system and this has to be stated clearly for marks to be awarded.

(iii) By conservation of momentum,
total initial momentum of system = total momentum of system at t_2 B1
$mv_0 = mv + Mv$

$$v = \frac{m}{m+M} v_0$$ A1

Comments:
Most students are able to get this part correct.

(c) Method 1:
By COM,
$4u(0.02c) + 0 = 4uv_a + 14u(0.005c)$ A1
final speed of alpha, $v_a = 0.0025$ c
Relative velocity of approach = $u_a - u_n = 0.02$ c A1
Relative velocity of separation = $v_n - v_a = 0.005$ c – (0.0025 c) = 0.0025 c A1
Since relative speed of approach is not equal to relative speed of separation, the interaction is not elastic.
OR
Method 2:
By COM,
$4u(0.02c) + 0 = 4uv_a + 14u(0.005c)$
final speed of alpha, $v_a = 0.0025$ c
Initial ke of alpha particle = $\frac{1}{2}(4u)(0.02c)^2 = 0.0008\ uc^2$
Final ke of nitrogen and alpha particle = $\frac{1}{2}(4u)(0.0025c)^2 + \frac{1}{2}(14u)(0.005c)^2$
$= 0.00019\ uc^2$
Since initial ke ≠ final ke, collision is not elastic

Comments:
There are many errors for this part:

1. Students are expected to deduce the mass of alpha particle and nitrogen from the number of nucleons by using the unified atomic mass.

2. Some students calculated the ke of alpha and nitrogen by substituting values. However, please note 1 u = 1.66×10^{-27} kg **NOT** 1.67×10^{-27} kg (which is the mass of proton). Marks will be deducted for substituting wrong values.

3. Some students showed that initial momentum of system not equal to final momentum of system, which is wrong concept.

4. When showing initial ke not equal to final ke, many students did not consider the final ke of alpha particle. Hence even if the conclusion is correct, no marks will be awarded as working is not conclusive.

5. Some students found the final velocity of alpha particle using relative velocity of approach = relative velocity of separation, and subsequently substituted the velocity found to find final ke and went on to conclude the collision is inelastic. This is not correct as when one uses relative velocity of approach = relative velocity of separation, one has assumed elastic head-on collision.